Register No: | | | | | | | | | | | | | |

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech. End Semester Theory Examinations, April / May 2024.

Sixth Semester

Information Technology

UIT2601 PATTERN RECOGNITION AND MACHINE LEARNING

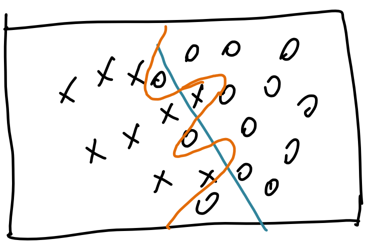
(Regulations 2021)

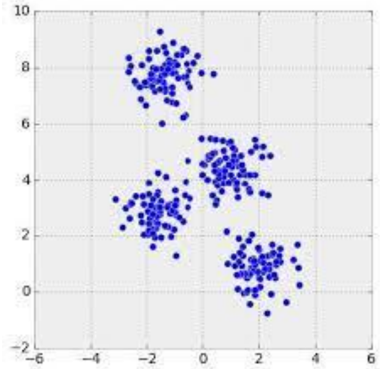
Time: **Three Hours** Maximum:**100 Marks**

K1: Remembering K2: Understanding K3: Applying K4: Analyzing K5: Evaluating

CO1	Explain machine learning algorithms and their limitations.
CO2	Apply common machine learning algorithms in practice and implement them for structured data.
CO3	Make use of deep learning algorithms for unstructured data.

Part – A (5 × 2 = 10Marks)

		KL	CO	PI
1.	Why can you not use mean square error as a cost function for logistic regression?	K3	CO1	
2.	Which of the following are regression (R) problems and which are classification (C) problems? (i) Given the no. of confirmed Dengue cases for the past 10 weeks, you are asked to find the probable no. of confirmed cases for the next week. (ii) Your client bank's lakhs of customers' data have been breached. You are asked to find out which of the clients' data has been breached. (iii) You have thousands of photos taken with your girl/boyfriend in your mobile phone. You want to hide these photos alone in a hidden folder, while retaining other photos as such before your hostile parent catches you. (iv) You work in a large retail store. You want to please your conservative boss by showing him the projection of sales of products for the next quarter.	K2	CO2	
3.	 The decision boundaries (orange and blue) of two algorithms are given in the figure. According to you, which algorithm would produce the orange, and which one would produce the blue decision boundary?	K2	CO1	

<table>
<tr><td>4.</td><td>If you are given the following dataset, where the training samples are given as $\{\boldsymbol{x}_i\}$, and asked to find to which group would a new data-point belong to, then which algorithm would you choose? Give the reason of your choice in one sentence.
</td><td>K3</td><td>CO2</td><td></td></tr>
<tr><td>5.</td><td>The following is the result of convolution. If you apply max-pooling, what would be the resultant matrix?
<table><tr><td>2</td><td>6</td><td>0</td><td>8</td></tr><tr><td>6</td><td>2</td><td>9</td><td>5</td></tr><tr><td>4</td><td>6</td><td>0</td><td>5</td></tr></table></td><td>K3</td><td>CO3</td><td></td></tr>
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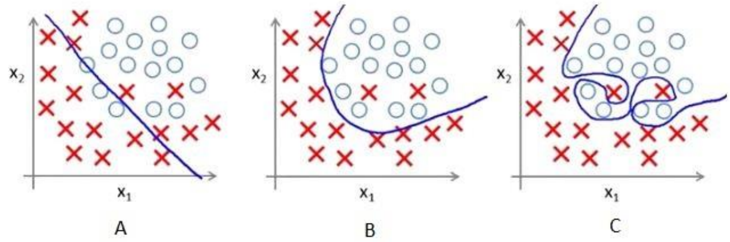
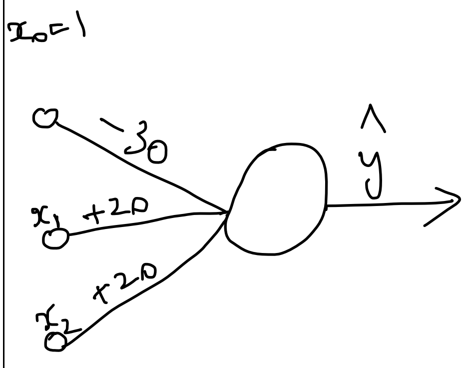
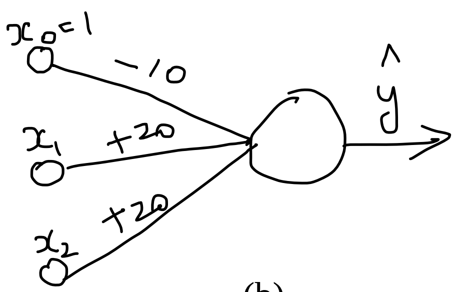
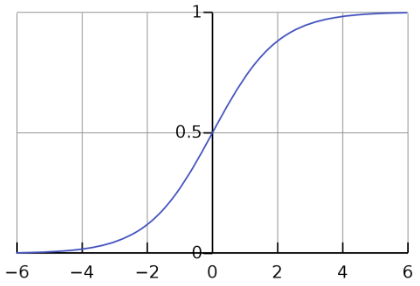
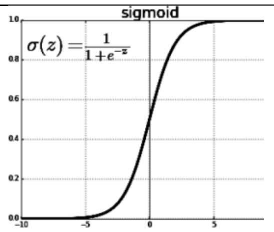
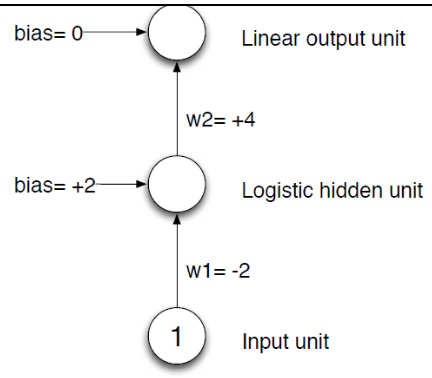
Part – B (5×6 = 30 Marks)

<table>
<tr><td></td><td></td><td>KL</td><td>CO</td><td>PI</td></tr>
<tr><td>6.</td><td>Which among the three plots in the figure below show that the decision boundary is the training data? Which has (a) overfitting (O), (b) underfitting (U), and, (c) which generalizes (G) best? Discuss with suitable examples.
</td><td>K2</td><td>CO1</td><td></td></tr>
<tr><td>7.</td><td>What logic gates do the following logistic regression units (a) and (b) represent? The sigmoid function (c) can be approximated from the given graph and need not be calculated.

(a)

(b)</td><td>K3</td><td>CO2</td><td></td></tr>
</table>

	 (c) sigmoid function			
8.	A neuron with 3 inputs has the weight vector **w** = $[0.1; 0.3; -0.2]^T$ and a bias = 0.35. The activation function is a sigmoid function. If the input vector is **x** = $[0.8; 0.6; 0.4]^T$ then what is the output of the neuron?	K3	CO2	
9.	A dataset comprises of datapoints (X) belonging to 2 classes – red and blue. A generative model is built for each class. If you are given P(X|Y=blue) = 0.02, P(X|Y=red) = 0.06, and if the prior probabilities of the two classes are taken as equal (i.e., P(Y=red) = P(Y=blue) = 0.5), to which class do you think the datapoint X would belong to?	K3	CO2	
10.	A 6x6 image, I and a 3x3 filter, F are as given below. I is: $\begin{array}{\|c\|c\|c\|c\|c\|c\|}\hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline 0 & 0 & 0 & 5 & 5 & 5 \\ \hline \end{array}$ And F is: $\begin{array}{\|c\|c\|c\|}\hline 1 & 0 & -1 \\ \hline 1 & 0 & -1 \\ \hline 1 & 0 & -1 \\ \hline \end{array}$ What is the result of the convolution operation, I * F?	K3	CO3	

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI
11.	 Consider the sigmoid activation function above: a) What would the gradient of the sigmoid be with respect to an input that is very large? b) Why might this be a problem for training neural networks? c) How does the ReLU activation ReLU(z) = max(0; z) solve this problem? d) How is tanh activation function different from sigmoid function? Illustrate with diagram. e) Will it help if you use tanh instead of ReLU? Give reason/s.	K2	CO1	

(Or)				
12.	 This is a very small neural network: it has one input unit, one hidden unit (logistic), and one output unit (linear). Let's consider one training case. For that training case, the input value is 1, and the target output value is 1. We're using the standard squared error loss function: E = (t-y)2 /2. The numbers in this question have been constructed in such a way that you don't need a calculator. a) What is the output of the hidden unit and the output unit, for this training case? b) What is the loss, for this training case? c) What is the derivative of the loss w.r.t. w2, for this training case? d) What is the derivative of the loss w.r.t. w1, for this training case?	K2	CO1	
13.	If you have to predict the no. of units sold of a particular mobile phone, given its cost (assume you have a training data set of 10000 different mobile phones), which machine learning algorithm would you use? Discuss with necessary equations.	K3	CO2	
(Or)				
14.	A cardiologist is interested in finding the influence of his patients' body mass and calories intake on heart attacks. He collects a database of 1000 patients, comprising their body mass, calories intake and whether they have had a heart attack or not. How would you help this cardiologist to predict heart attacks on new patients using a machine learning algorithm? What is this problem - classification or regression? Write only the steps of the algorithm, including the cost function and the weight adjustment.	K3	CO2	
15.	The Arrhythmia (rhythm of heart) dataset has 279 attributes and 452 instances. A student ran multiple machine learning algorithms on the dataset and, for each algorithm, achieved about 75% accuracy in prediction. Propose two algorithms that will minimize the variance and improve the overall performance over the base systems. Discuss with suitable equations and illustrations.	K3	CO2	
(Or)				

16.	A pulmonologist wants you to automate the process of pneumonia detection using chest X-rays. He has collected data pertaining to pneumonia from the X-rays of 5000 patients. He looks at 2 parameters - presence of white spots or infiltrates in the lung, and presence of fluid surrounding the lungs - as indication of pneumonia. As a machine learning engineer, what problem do you identify this as, and what is the solution you propose for this automation? Discuss the appropriate algorithm in detail.	K3	CO2	
17.	For a given bank customer dataset, you try different ML algorithms, and each time the accuracy lies between 65% to 70%. How can you improve the accuracy by using an ensemble of these classifiers? Discuss your solution using bagging, boosting, and stacking.	K3	CO2	
	(Or)			
18.	You get a dataset that has images of CT scans of people, without the diagnosis given. Since you know that some of these scans could be related to COVID symptoms, some related to yet another symptom, and some could be normal, etc. You are asked to group these images. Discuss GMM-EM algorithm that would do this for you.	K3	CO2	
19.	An image dataset is given to you. One of images looks like this: input image  You are asked to classify them into one of ten classes (digits 0 to 9). Design a convolutional neural network for this task. Illustrate clearly the # layers used, the activation functions used, how the dimensions change with each layer, etc. Explain your design.	K4	CO3	
	(Or)			
20.	You are a consultant for a healthcare company. They provide you with clinical notes of 2 million patients, each clinical note mentioning the underlying illnesses and the diagnosis made by the doctor. A clinical note looks something like this:	K4	CO3	

	<u>History</u> ROS: no change in bowel/urinary habits Meds: no Rx or OTC. All: NKDA. FH: mother - schizophrenia PMH: asthma, good control. No surgeries, traumas or hospital. SxH: sex. active with multiple F and M partners, inconsistent use of condoms, no h/o STDs SH: NO cig/eoth. Uses PCP and ecstasy x 1y, once/week, last intake yesterday. College student. **<u>Physical Examination</u>** Pt is in NAD. Speech fluent, talkative, mood euphoric, affect c/w mood, behavior inappropriate. Cooperative. Appearance disheveled. VS: WNL HEENT: EOMI, PERRLA. Neck: NL Thyroid Gland For a particular analytical application, you are required to convert these words to vector representations. How would you use RNN to accomplish this?			